



NSW Education Standards Authority

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Centre Number

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Student Number

2025 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Standard 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page

Total marks:
80

Section I – 10 marks (pages 2–7)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 70 marks (pages 9–32)

- Attempt Questions 11–28
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

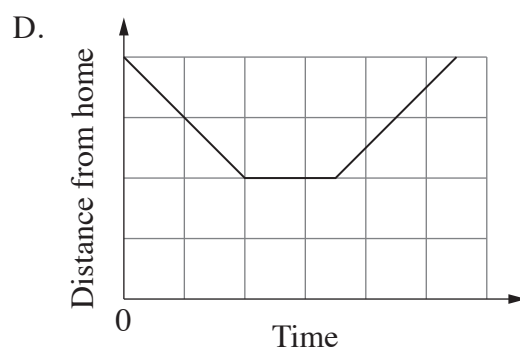
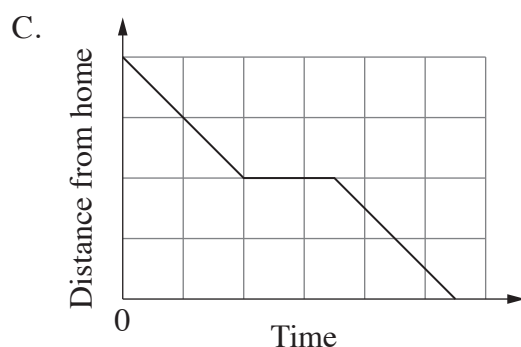
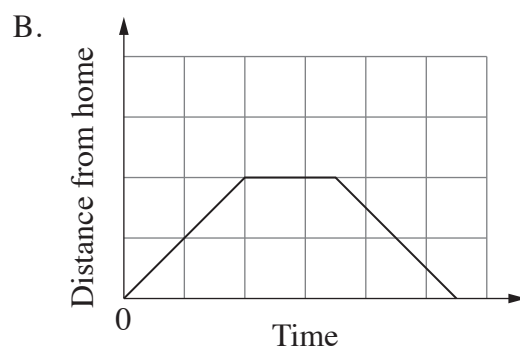
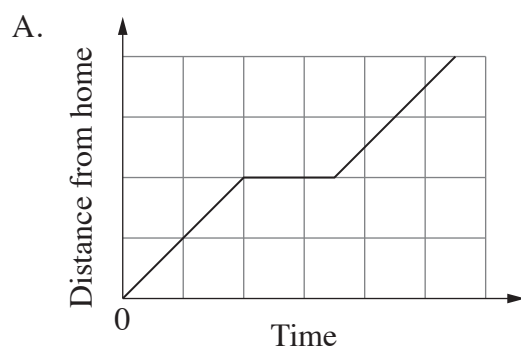
Use the multiple-choice answer sheet for Questions 1–10.

1 Which of the following could be classified as discrete data?

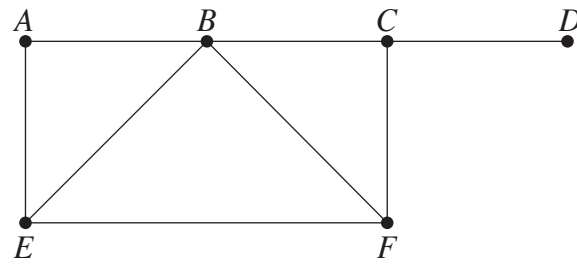
- A. Colour of a car
- B. Time taken to swim 200 m
- C. Temperature of an ice block
- D. Number of students in a class

2 Mario drives from his home to his friend's house. He watches a movie at his friend's house and then drives home.

Which distance–time graph best represents Mario's complete journey?



- 3 Consider the network diagram.

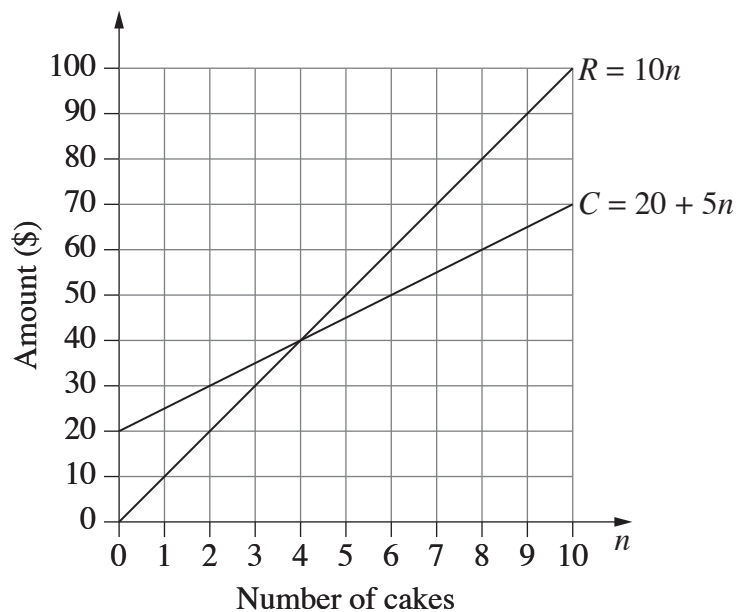


Which vertex has degree 4?

- A. *A*
 - B. *B*
 - C. *C*
 - D. *D*
- 4 What is 6 280 000 in standard form?
- A. 628×10^4
 - B. 62.8×10^5
 - C. 6.28×10^6
 - D. 0.628×10^7

- 5 A baker makes and sells cakes.

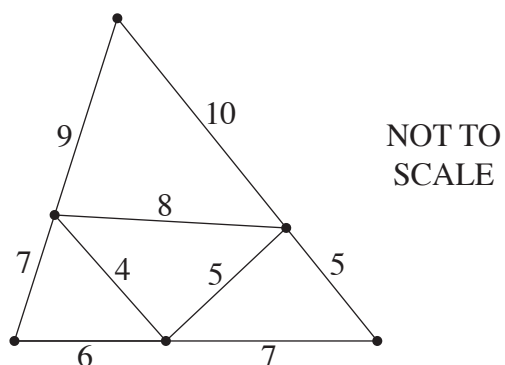
The straight-line graphs represent cost (C) and revenue (R) in dollars, and n is the number of cakes.



What profit will the baker make by selling 6 cakes?

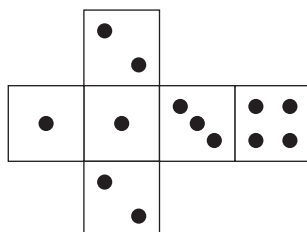
- A. \$10
- B. \$20
- C. \$40
- D. \$60

- 6 The network shows the distances, in kilometres, along a series of roads that connect towns.



What is the value of the largest weighted edge included in the minimum spanning tree for this network?

- A. 7
B. 8
C. 9
D. 10
- 7 A biased die is made from this net.

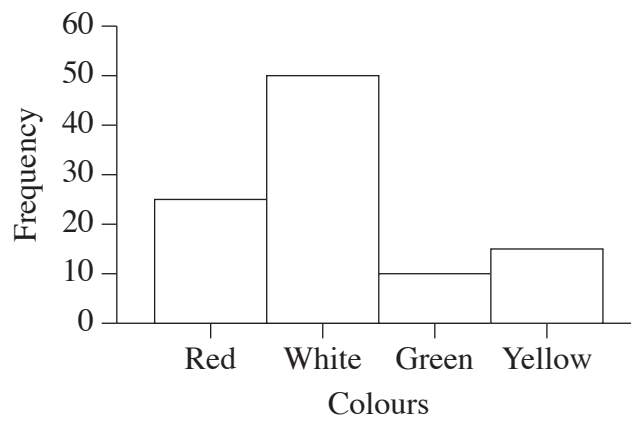


The die is rolled once.

What is the probability of rolling a 2?

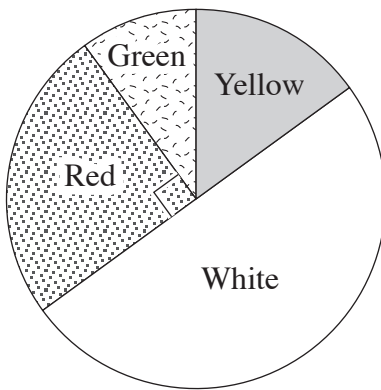
- A. $\frac{1}{6}$
B. $\frac{1}{4}$
C. $\frac{1}{3}$
D. $\frac{1}{2}$

- 8 A spinner made up of 4 colours is spun 100 times. The frequency histogram shows the results.

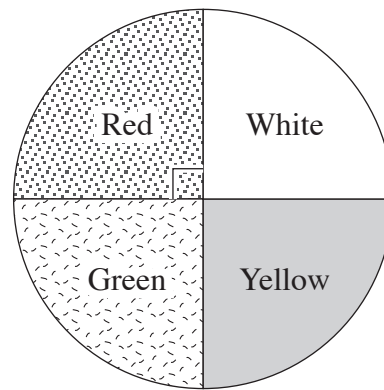


Which of these spinners is most likely to give the results shown?

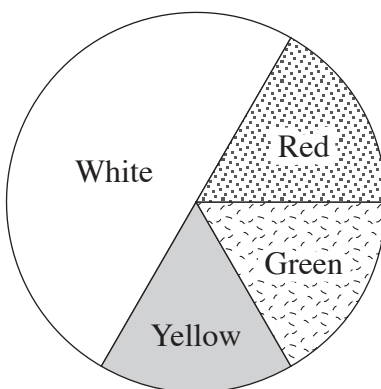
A.



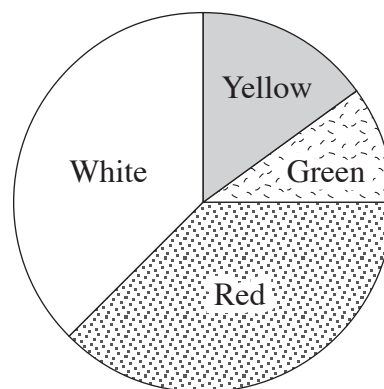
B.



C.



D.



- 9** An amount of \$90 000 is invested at 4% per annum, compounded quarterly.

Which expression gives the value of this investment, in dollars, after 6 years?

- A. $90\,000(1 + 0.04)^6$
 - B. $90\,000(1 + 0.04)^{24}$
 - C. $90\,000(1 + 0.01)^6$
 - D. $90\,000(1 + 0.01)^{24}$
- 10** The ratio of the dimensions of a model car to the dimensions of an actual car is 1 : 64.
The actual car has a length of 4.9 m.

What is the length of the model car in cm, correct to 1 decimal place?

- A. 3.1
- B. 7.7
- C. 13.1
- D. 59.1

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Centre Number

Mathematics Standard 1

Section II Answer Booklet

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Student Number

70 marks

Attempt Questions 11–28

Allow about 1 hour and 45 minutes for this section

Instructions

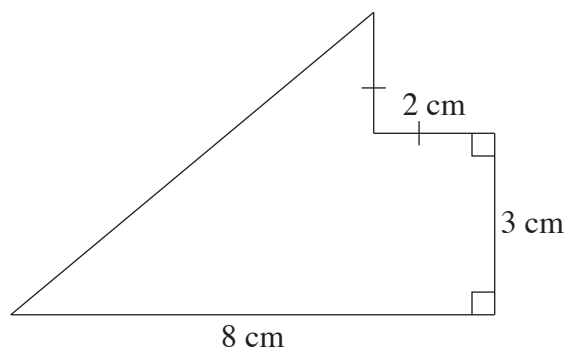
- Write your Centre Number and Student Number at the top of this page
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response
- Your responses should include relevant mathematical reasoning and/or calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering

Please turn over

Question 11 (3 marks)

Consider the composite shape shown.

3



What is the area of the shape in cm^2 ?

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Question 12 (2 marks)

Vijay's heart rate before and after his morning run is shown.

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<i>Before run</i>	72 beats per minute
<i>After run</i>	126 beats per minute

What is the percentage increase in Vijay's heart rate?

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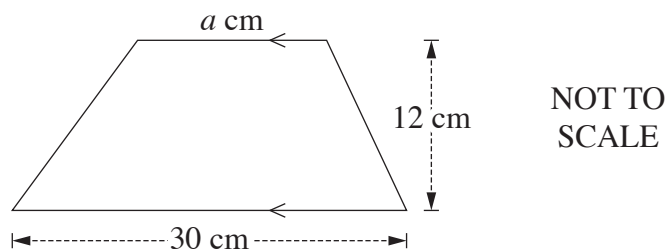
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Question 13 (3 marks)

A trapezium is shown.



- (a) Write an expression for the area of this trapezium.

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- (b) Find the value of a , given that the area of the trapezium is 330 cm^2 .

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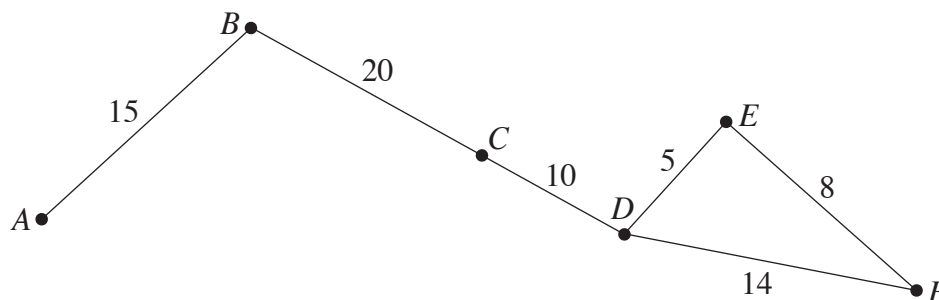
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Question 14 (5 marks)

The time, in minutes, it takes to travel by road between six towns is recorded and shown in the network diagram below.



- (a) In this network the shortest path corresponds to the minimum travel time. 2

What is the minimum travel time between towns *A* and *F*, and what is the corresponding path?

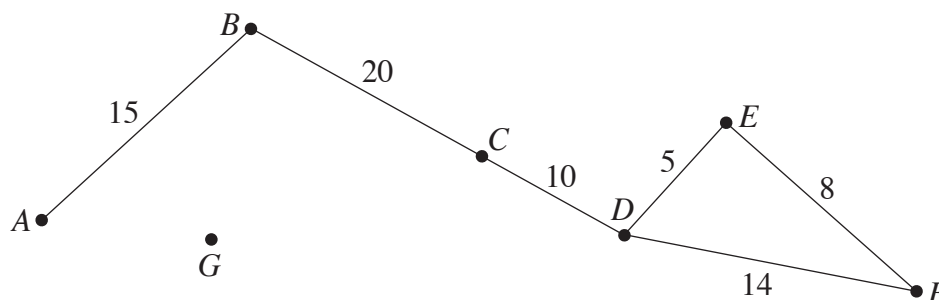
Minimum travel time:

Path:

New roads are built to connect a town *G* to towns *A* and *D*. The table gives the times it takes to travel by the new roads.

<i>Town</i>	<i>Time (minutes)</i>	<i>Town</i>
<i>A</i>	8	<i>G</i>
<i>G</i>	22	<i>D</i>

- (b) Add the new roads and times to the network diagram below. 2



- (c) Explain whether the path in part (a) is still the shortest path from *A* to *F* after the new roads are added. 1

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Question 15 (8 marks)

A researcher is using the statistical investigation process to investigate a possible relationship between average number of minutes per day a person spends watching television, and the average number of minutes per day the person spends exercising.

- (a) State the statistical question being posed.

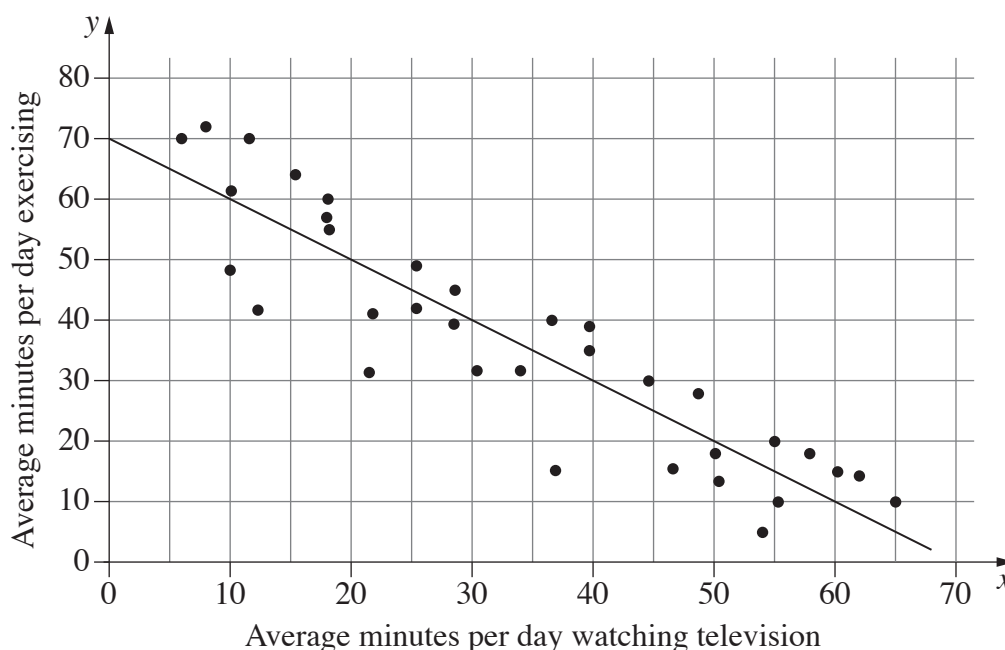
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Participants were asked to record the number of minutes they spent watching television each day and the number of minutes they spent exercising each day. The averages for each participant were recorded and graphed, and a line of best fit was included.



- (b) From the graph, identify the dependent variable.

1

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Question 15 continues on page 15

Question 15 (continued)

- (c) Describe the bivariate dataset in terms of its form and direction. 2

Form:

Direction:

- (d) The points (0, 70) and (60, 10) lie on the line of best fit. By first plotting these points on the graph, find the gradient and the y-intercept of the line of best fit. 3

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- (e) Explain why it is NOT appropriate to extrapolate the line of best fit to predict the average number of minutes of exercise per day for someone who watches an average of 2 hours of television per day. 1

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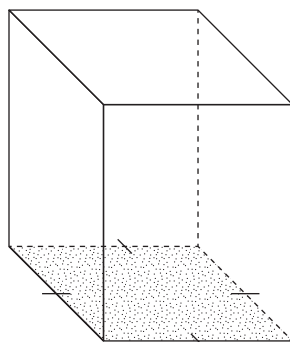
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End of Question 15

Question 16 (3 marks)

The mass (M) of a box with a square base, in grams, is directly proportional to the area of its base, in cm^2 .

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NOT TO
SCALE

A box with a square base of side length 5 cm has a mass of 500 g.

What is the mass of a similar box with a square base of side length 3 cm?

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Question 17 (2 marks)

Paint is sold in two sizes at a local shop.

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- Four-litre cans at \$90 per can
- Ten-litre cans at \$205 per can

Mina needs to purchase 80 litres of paint.

Calculate the amount of money Mina will save by purchasing only ten-litre cans of paint rather than only four-litre cans of paint.

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Question 18 (3 marks)

Ramon took 1.25 hours to travel the length of a freeway. The length of the freeway is 100 km.

(a) Ramon's total travel time was made up of:

2

- 20 minutes to travel to the start of the freeway
- the time taken to travel the length of the freeway
- 35 minutes after exiting the freeway to get to his destination.

What was the total time Ramon spent travelling? Give your answer in hours and minutes.

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(b) What was Ramon's average speed on the freeway?

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Question 19 (4 marks)

At the end of the 2024–2025 financial year, Alex’s taxable income was \$148 600.

- (a) The table shows the income tax rate for Australian residents for the 2024–2025 financial year. **3**

<i>Taxable income</i>	<i>Tax on this income</i>
0 – \$18 200	Nil
\$18 201 – \$45 000	16 cents for each \$1 over \$18 200
\$45 001 – \$135 000	\$4288 plus 30 cents for each \$1 over \$45 000
\$135 001 – \$190 000	\$31 288 plus 37 cents for each \$1 over \$135 000
\$190 001 and over	\$51 638 plus 45 cents for each \$1 over \$190 000

Using the table, calculate Alex’s tax payable.

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- (b) The Medicare levy is 2% of taxable income. **1**

Calculate the Medicare levy payable by Alex.

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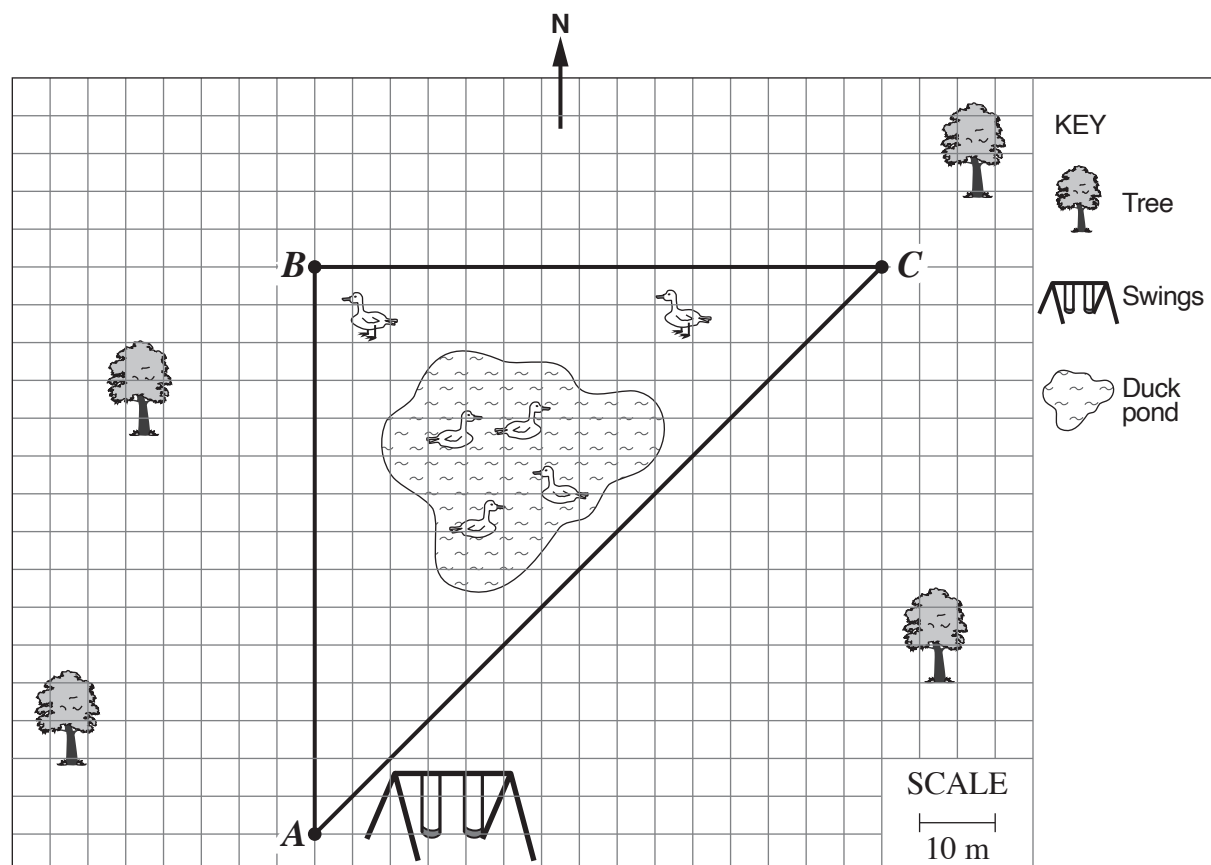
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Questions 11–19 are worth 33 marks in total

Question 20 (7 marks)

A map of a park containing a duck pond is shown.

A fence is built passing through the points A , B and C around the duck pond.



- (a) Using the scale provided on the map, calculate the length of the fence AB .

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Question 20 continues on page 21

Question 20 (continued)

- (b) The length of AB is equal to the length of BC .

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Use Pythagoras' theorem to calculate the length of AC in metres. Give your answer correct to 3 significant figures.

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- (c) What is the true bearing of point A from point C ?

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End of Question 20

Question 21 (3 marks)

A credit card has an interest-free period of 45 days from and including the date of purchase. Interest is charged on purchases made, compounding daily at a rate of 13.74% per annum, from and including the day following the interest-free period.

3

Concert tickets were purchased for a total of \$392 using this credit card.

Full payment was made on the 68th day from the date of purchase. There were no other purchases on this credit card.

What was the total interest charged when the account was paid in full?

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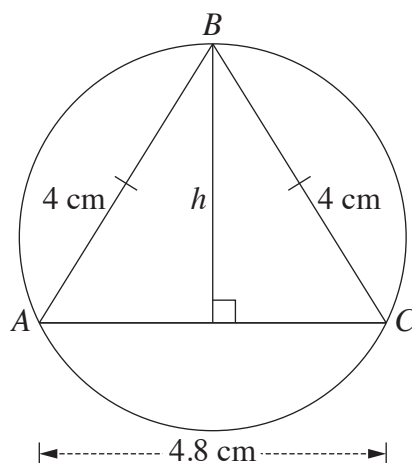
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Question 22 (4 marks)

An isosceles triangle is drawn inside a circle as shown. The base of the triangle is 4.8 cm long, the length of other sides is 4 cm and the height is h cm.



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SCALE

- (a) Calculate the height, h , of triangle ABC .

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- (b) The area of triangle ABC is 7.68 cm^2 .

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The radius of the circle is 2.5 cm.

Express the area of triangle ABC as a percentage of the area of the circle, correct to 1 decimal place.

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Question 23 (2 marks)

It costs \$465 to register a passenger car and \$350 to register a motorcycle.

2

Let P = the number of passenger cars, and

B = the number of motorcycles.

Write TWO linear equations that represent the relationships below.

- There are 11 times as many passenger cars as motorcycles.
- The total registration fees for passenger cars and motorcycles is \$494 million.

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Question 24 (5 marks)

A used car has a sale price of \$24 200. In addition to the sale price, the following costs are charged:

5

- transfer of registration \$50
- stamp duty which is calculated at \$3 for every \$100, or part thereof, of the sale price.

Kat borrows the total amount to be paid for the car, including transfer of registration and stamp duty. Simple interest at the rate of 6.8% per annum is charged on the loan. The loan is to be repaid in equal monthly repayments over 3 years.

Calculate Kat's monthly repayment.

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Question 25 (4 marks)

Bobbie plans to invest \$25 000 for 10 years and is offered two investment options.

4

Option A: Earns interest at a rate of 5% per annum, compounding monthly.

Option B: Earns simple interest at a rate of 8% per annum.

Which investment option will provide Bobbie with the best return value at the end of 10 years? Justify your answer with calculations.

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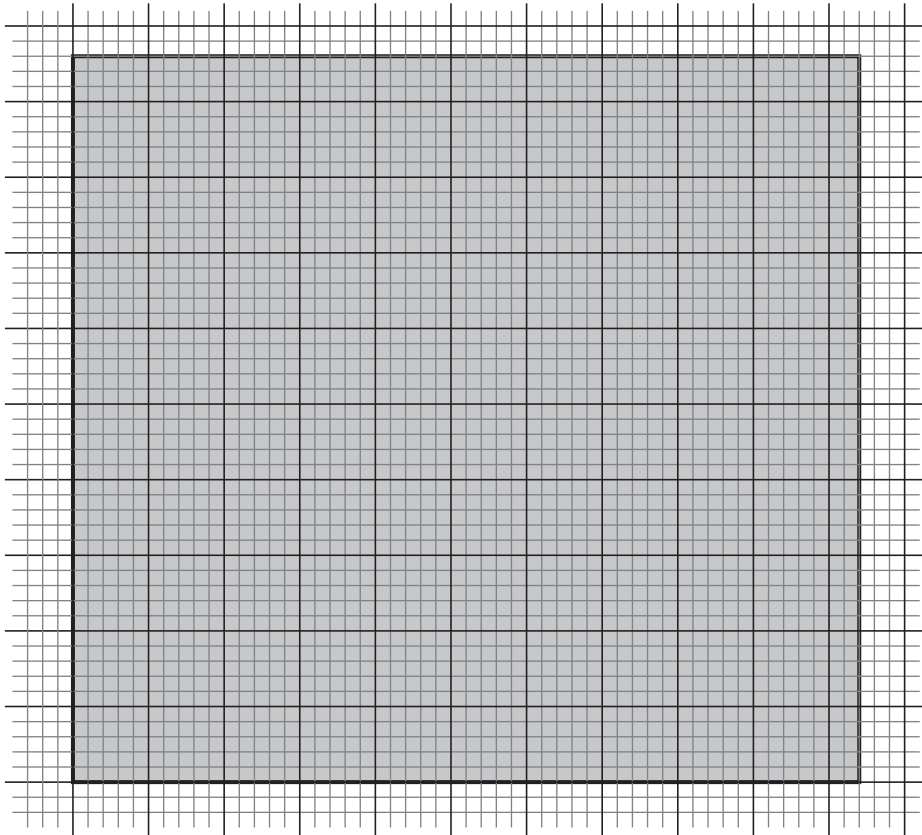
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Question 26 (4 marks)

A scale of 1 : 50 is used to draw a rectangular area on a 2 mm grid as shown. The actual rectangular area is to be tiled.

4



The tiles cost \$150 per square metre and the tiler orders 15% extra tiles to allow for cutting and breakage.

The tiler charges \$90 per hour and will take 20 hours to complete the tiling.

Calculate the total cost of the tiles and tiling. Give your answer to the nearest dollar.

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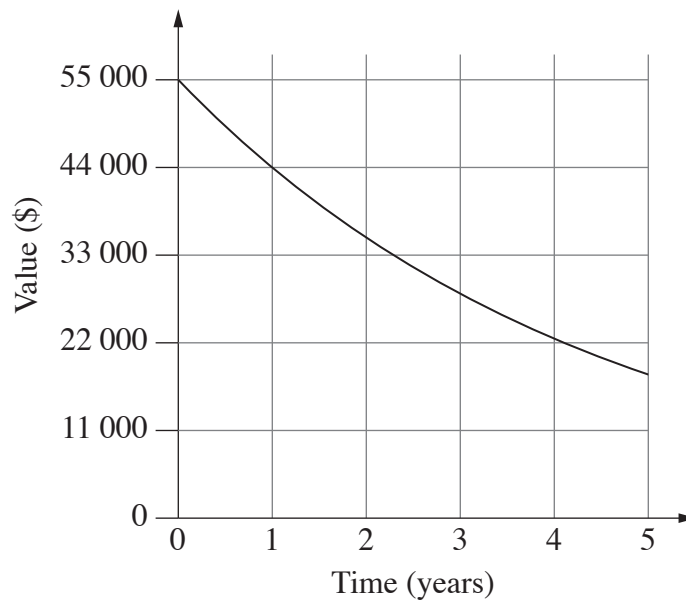
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Question 27 (4 marks)

The graph shows the salvage value of a car over 5 years.

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The salvage values are based on the declining-balance method.

By what amount will the car's value depreciate during the 10th year?

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Question 28 (4 marks)

The table provides information about a \$2 coin and a \$5 note.

<i>Coin/note</i>	<i>Quantity needed to make \$1000</i>	<i>Mass of this number of notes/coins (kg)</i>
\$2	500	3.3
\$5	200	0.157

- (a) Calculate the mass of a \$2 coin in grams, correct to 1 decimal place. 2

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- (b) Suppose the \$2 coin is to be replaced with a note that has the same mass as a \$5 note. 2

What is the mass of \$1000 in \$2 notes in grams? Give your answer correct to the nearest gram.

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End of paper

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Mathematics Standard 1

Mathematics Standard 2

REFERENCE SHEET

Measurement

Limits of accuracy

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

An outlier is a score

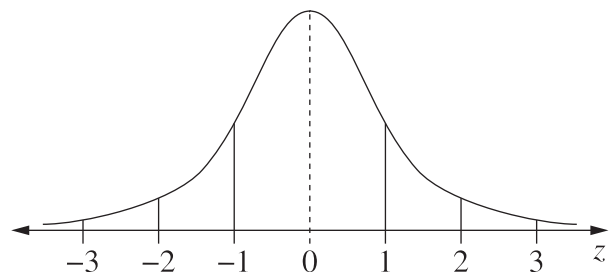
less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \mu}{\sigma}$$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

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